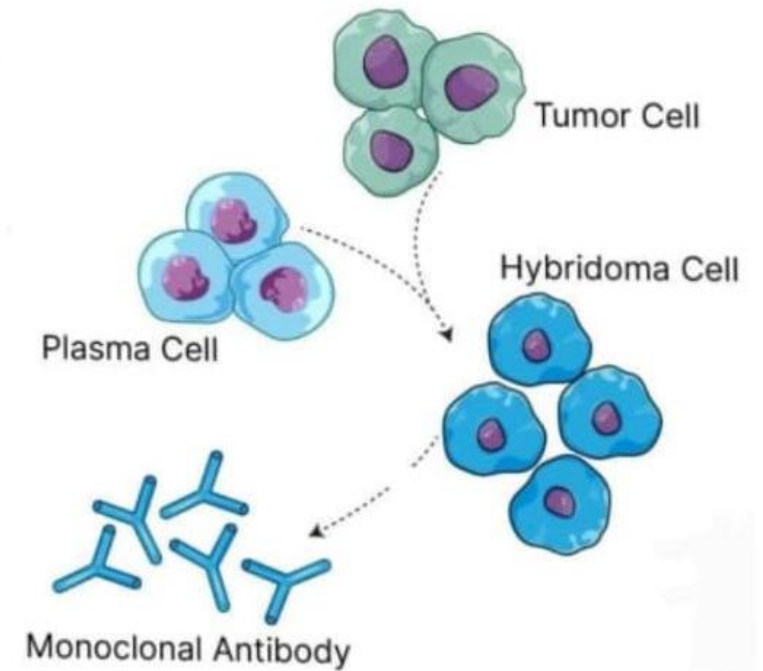


Introduction of hybridoma



- 
- **Antigen:** any foreign substance could or couldn't stimulate immune response.
 - **Immunogen:** a substance that stimulate immune responses.
 - All immunogens are Antigens, but not all Antigens are immunogens.
 - **Epitopes:** specific regions of Ag where Antibody or TCR (T cell Receptor) bind.

- 
- If Foreign protein contain 5 different epitopes inject to Mouse→ five different types of B cells were produced
 - Epitope 1→ B cell 1→Antibody against epitope 1
 - Epitope 2→B cells 2 →Antibody against epitope 2
 - Epitope 3→B cell 3 →Antibody against epitope 3
 - Epitope 4→ B cells → Antibody against epitope 4
 - Epitope 5→B cell 5→ Antibody against epitope 5

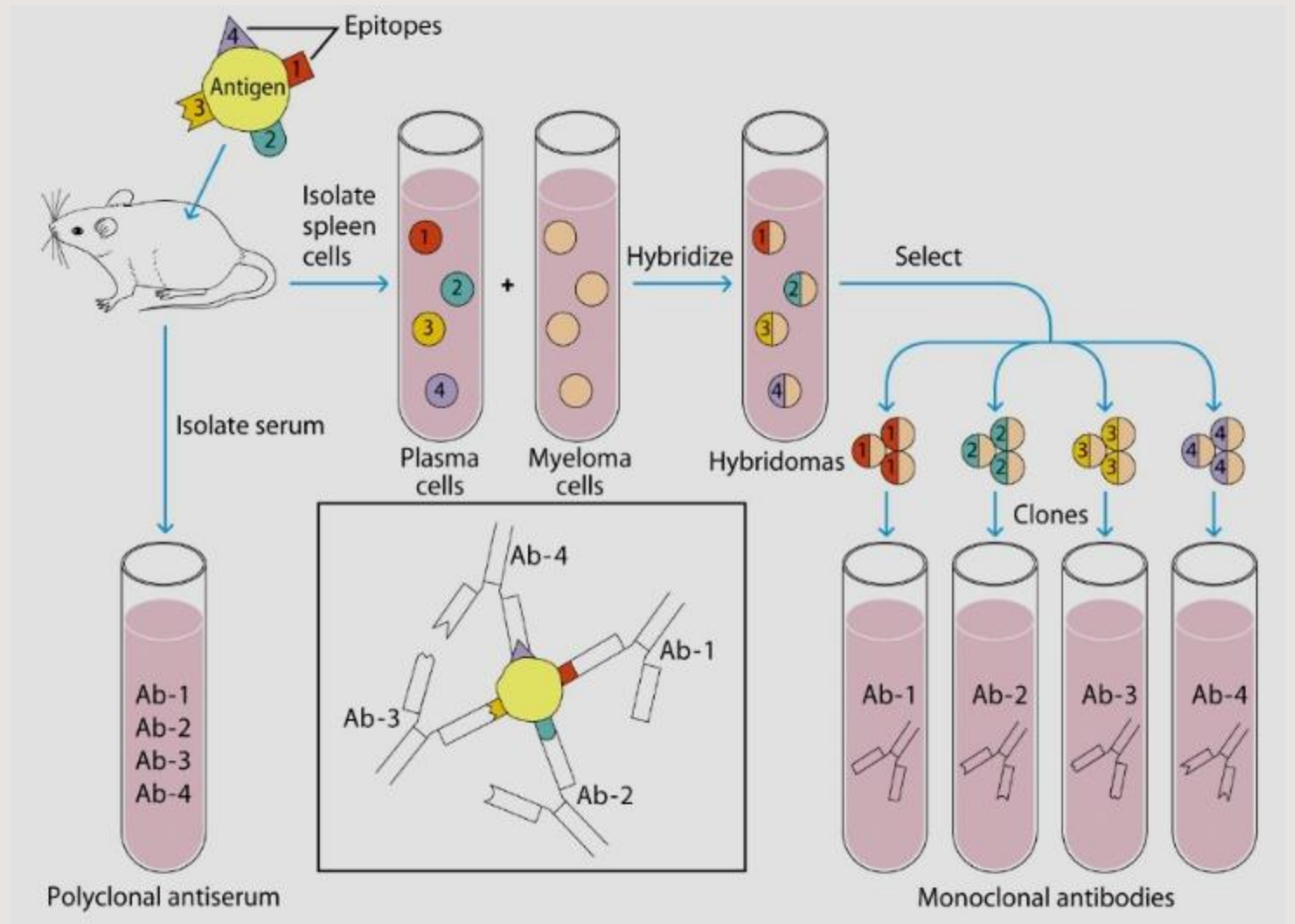
- Five types of B cells exist in secondary lymphoid spleen , lymph nodes,...

While Antibodies exist in serum.

Isolation blood from injected mouse → mixture of Antibodies → Polyclonal Antibodies.

Polyclonal Antibodies → mixture of Antibodies produced by different B cells clones.

Isolate each B cell alone → produce monoclonal antibodies.



- 
- Monoclonal Antibodies→ solution of Antibodies produce by single B cell type.
 - Cross reaction→common epitopes between two Antigens.
 - In diagnosis, cross reaction cause to mis diagnostic



1-Salmonellosis→ cause by Salmonella, Salmonella has epitopes, A, B and C.

- Isolation serum from patient→ Polyclonal Antibodies against epitope A, B and C.

2- In Brucellosis→ cause by Brucella, Brucella has epitopes A, B and D.

- Isolation serum from patient→ Polyclonal Antibodies against epitopes A, B and D.

- **In diagnosis**

- Infection → Antibodies against infection → isolation of Antibodies → using kit (contain the Ag from infection).

- If the reaction occur person infected, if not person not infected.

- Common epitopes between Salmonella and Brucella → mis diagnostic.

- 
- Isolation of Ab from Brucellosis → positive result in both Brucellosis and Salmonellosis.
 - Isolation Ab from Salmonellosis → positive result in both Brucellosis and Salmonellosis.
 - To solve this problem, using Kits contain unique epitopes not common epitopes.
 - Kits contain only epitope C → only react with Salmonellosis serum.
 - Kits contain only epitope D → only react with Brucellosis.

	POLYCLONAL	MONOCLONAL
ADVANTAGES	• Inexpensive to produce • High affinity • Recognize multiple epitopes (generally provides more robust detection) • Polyclonal antibodies are often preferred for detection of denatured proteins • Higher tolerance for differences in antigen (i.e. glycosylation of proteins)	• Constant and renewable source • Consistency between lots • Less background relative to polyclonal antibodies • Homogeneity ensures reproducible results • Specificity of monoclonal antibodies make them extremely efficient for binding of antigen within a mixture of related molecules
DISADVANTAGES	• Prone to batch to batch variability • They produce large amounts of non-specific antibodies which can result in a background signal in some applications • Multiple epitopes make it important to check for cross reactivity	• Monoclonal antibodies may be too specific (e.g. less likely to detect across a range of species) • Cover only one epitope

Challenges of hybridoma technology :

- 1) This is a time-consuming method, requiring 6 to 9 months.
- 2) The method is quite expensive and requires considerable effort in production.
- 3) This method is not suitable for producing antibodies against small peptides and fragment antigens.
- 4) Hybridoma culture suffers from a high risk of contamination [10, 18].
- 5) This system of antibody production is now developed for only mice and rats and researchers are working continuously to develop antibodies of human origin [19].



6) The viable efficiency of cells is quite low. More than 99% of the cells die during the process of cell fusion reducing the efficiency of the the method and also reducing the variety of helpful helpful lantibodies antibodies which will be made against a specific substance.

7) If monoclonal antibodies are generated against a single antigenic determinant, they do not show cross-reactivity with other antigenic determinants.

Retroviruses are a common incidence within mammalian chromosomes. Generally, animals like mice that are used in the production of monoclonal antibodies could carry many viruses such as viscus virus, retrovirus, reovirus, herpes virus, and thymic virus, leading to cross- contamination n or infection in humans (18). This poses a major threat for cross disease transfer from mice or rats to the human.



8)Despite purification, there is no guarantee that monoclonal antibodies made using the hybridoma technique are virus-free. The fusion of human lymphocyte and mouse myeloma cells may result in the production of unstable fused cells.

9)In humans, there are no stable myeloma cells, suitable for the process of antibody production that can be used to substitute mouse myeloma cells.



Advantages of hybridoma technology:

- 1) Hybridoma technology produces highly pure and specific antibodies (monoclonal antibodies).
- 2) This method is highly reproducible and scalable [1].
- 3) This method provides an unlimited production of monoclonal antibodies.
- 4) It can be used to perform highly sensitive and specific assays.



5) There is no need to maintain the animal in the laboratory for the production of antibodies [in vitro method] [11].

6) In this method, the purity of antigen or immunogen is not a prerequisite.

7) The selection method is useful in the identification of the right clones against a specific antigen.

8) In vivo production of antibodies ensures the formation of a mixture of variable and constant domains. The generated antibodies have a high affinity towards the epitope of a targeted substance [6].



9) This method is not labor-intensive; in vitro antibody generation techniques require the use of immune libraries.

10) Antibody reliability is vital to each analysis and assay development and is one of the key features of hybridoma technology. Once hybridoma cells become stable, these offer limitless production of cost-effective, homogenized antibodies [1].

11) The perfect tool for research, in various fields such as toxicology, animal biotechnology, medicine, pharmacology, etc.

12) Monoclonal antibodies are widely employed in diagnostic and therapeutic procedures [17].

13) It is used in various chemotherapeutic regimens to treat various cancer types.

14) Used widely in research for the production of vaccines.



This technology offers numerous advantages:

- 1) Precise antigen targeting
- 2) A never-ending supply of consistent antibodies
- 3) High sensitivity and specificity for use in biological assays
- 4) Elimination of the need for animal models (in vitro method)
- 5) Utilization in therapeutic and diagnostic treatments, vaccine creation, and chemotherapy



Nevertheless, the technology also has a few limitations :

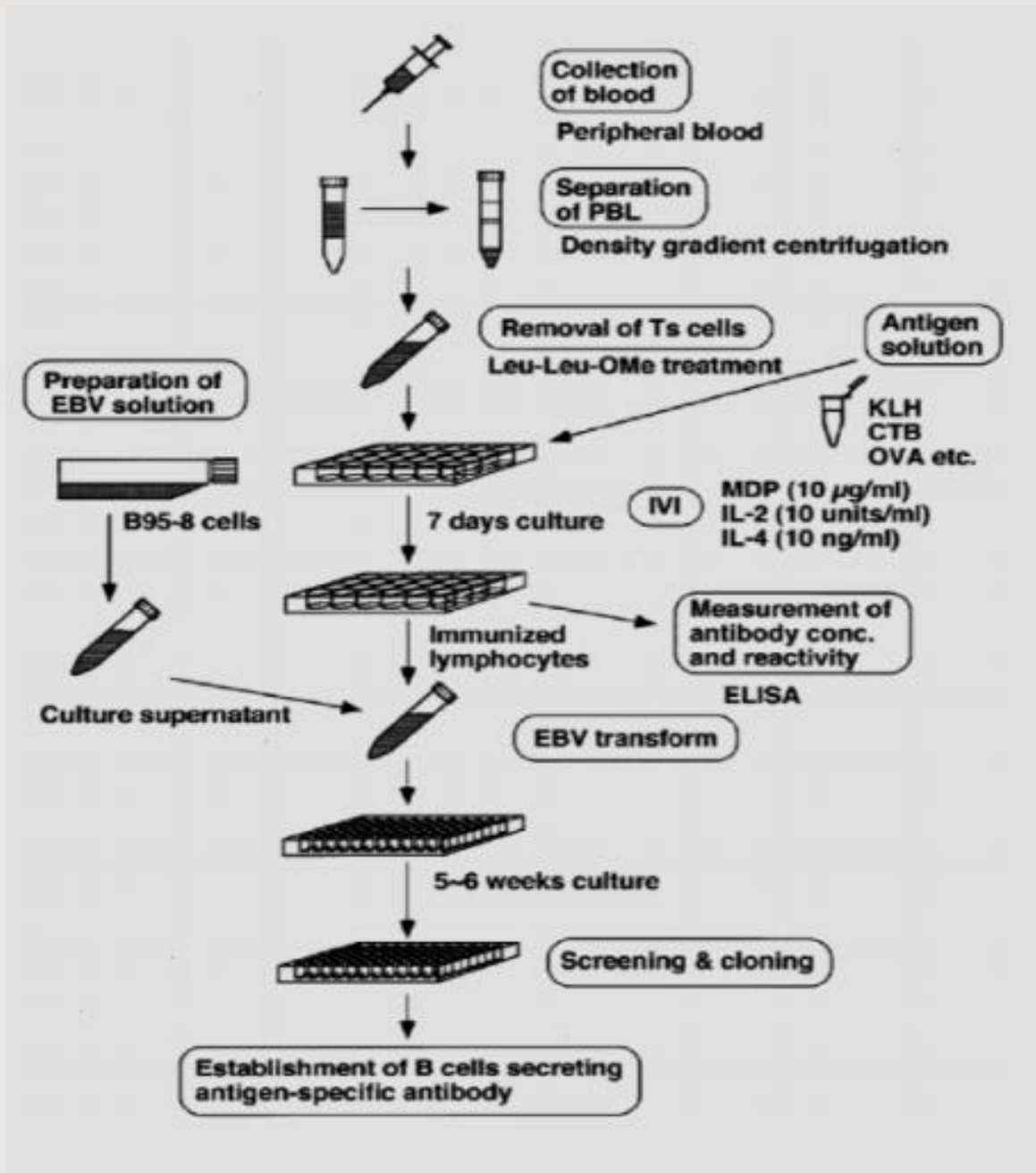
- 1) Long production time
- 2) Resource-intensive and expensive workflow
- 3) Not suitable for generating short peptides and fragment antigens
- 4) Susceptibility to contamination and poor cell viability
- 5) Risk of virus contamination and disease transmission
- 6) Absence of stable myeloma cells for human antibody production

The principle of Monoclonal production

- Ag injection of mouse → specific B cells in Spleen against Ag → specific Antibodies.
- Hybridoma cells → a cells produce by fusion between Plasma cells (B cells that secrete Antibodies) and Myeloma cells (Cancer B cells).

Hybridoma cells

- Immortal → grow indefinitely
- Antibody producing cells



Applications of Mabs

1) Diagnostic application:

- Diagnostic tests based on the use of Mabs as reagents are routinely used in radioimmunoassay (RIA) and enzyme-linked immunosorbent assays (ELISA) in the laboratory.
- These assays measure the circulating concentrations of hormones (insulin, human chorionic gonadotropin, growth hormone, progesterone, thyroxine, triiodothyronine, thyroid stimulating hormone, gastrin, renin),
- it is now possible to do the early diagnosis of the following conditions/diseases: pregnancy, cancer, hormonal disorders, infectious diseases, cardiovascular diseases (atherosclerosis, DVT, myocardial infarction, immunohistopathology and immunohaematopoietic malignancies

2) Therapeutic Applications:

- Monoclonal antibodies have a wide range of therapeutic applications. Mabs are used in the treatment of cancer, transplantation of bone marrow and organs, autoimmune diseases, cardiovascular diseases and infectious diseases.
- In 2018, a record number (12) of antibodies that treat a wide variety of diseases were granted a first approval in either the European Union (EU) or United States (US).

- 
- November 2018, 4 Ab therapeutics (sacituzumab govitecan, ravulizumab, risankizumab, romosozumab) were being considered for their first marketing approval in the EU or US.
 - Finally, approximately half (18 of 33) of the late-stage pipeline of Ab therapeutics for cancer are immune checkpoint modulators or Ab-drug conjugates are found. Of these, 7 are being evaluated in clinical studies with primary completion dates in late 2018 and in 2019
 - On October 14, 2020, FDA approved the triple Ab cocktail of atoltivimab, maftivimab, and odesivimab-ebgn (Inmaze) for the treatment for (Ebola virus) infection in adult and pediatric patients. Atoltivimab, maftivimab, and odesivimab are IgG1 antibodies that bind glycoprotein on the surface of Ebola virus, thereby blocking attachment and entry of the virus into cells.
 - As of mid-November 2022, 12 Mab therapeutics had been granted first approvals

3. Protein Purification and identification:

- Monoclonal antibodies can be produced for any protein. And the produced Mab can be conveniently used for the purification of the protein against which it was raised.
- Mabs columns can be prepared by coupling them to cyanogen bromide activated Sepharose (chromatographic matrix). The immobilized MAbs in this manner are very useful for the purification of proteins by immunoaffinity method.

Limitations for direct use of Mabs in cancer:

1. The Mabs produced in mice and directly used for therapeutic purposes may lead to the development of anti-mouse antibodies and hypersensitivity reactions.
2. All the cancer cells may not carry the same antigen for which Mab has been produced. Thus, Mabs may not be attached to some cancer cells at all.
3. The free antigens (of target cells) present in the circulation may bind to Mabs and prevent them from their action on the target cells.